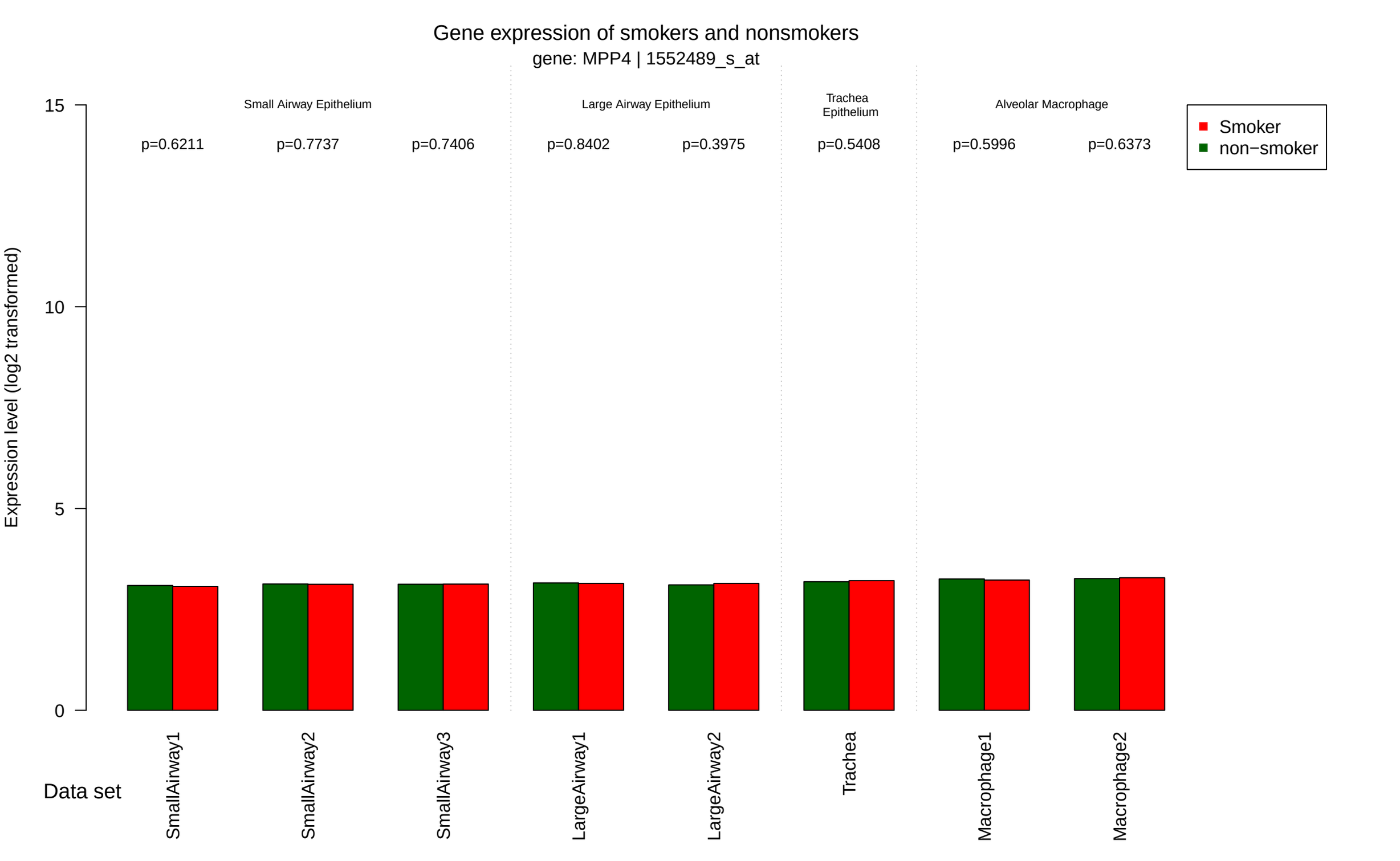
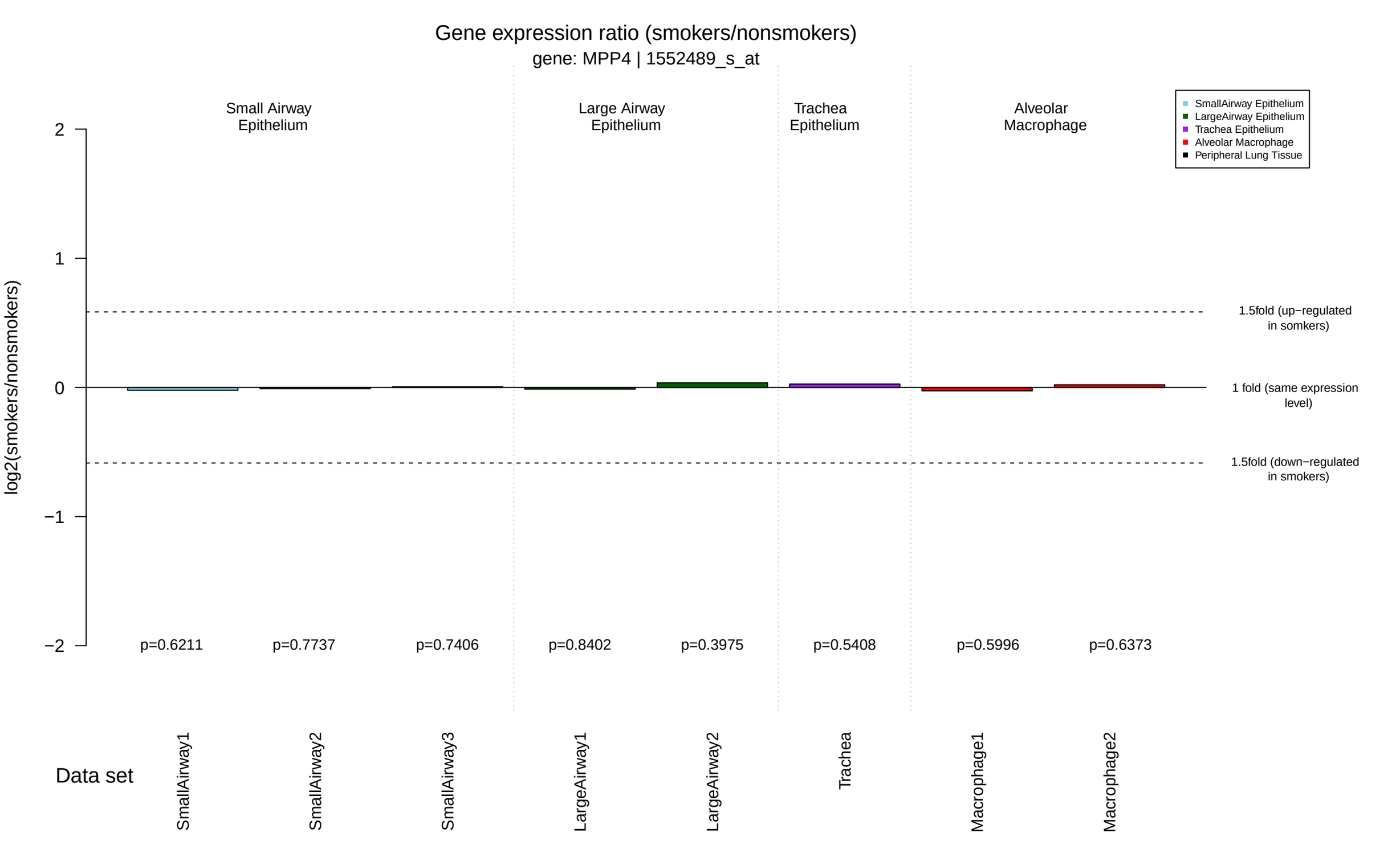
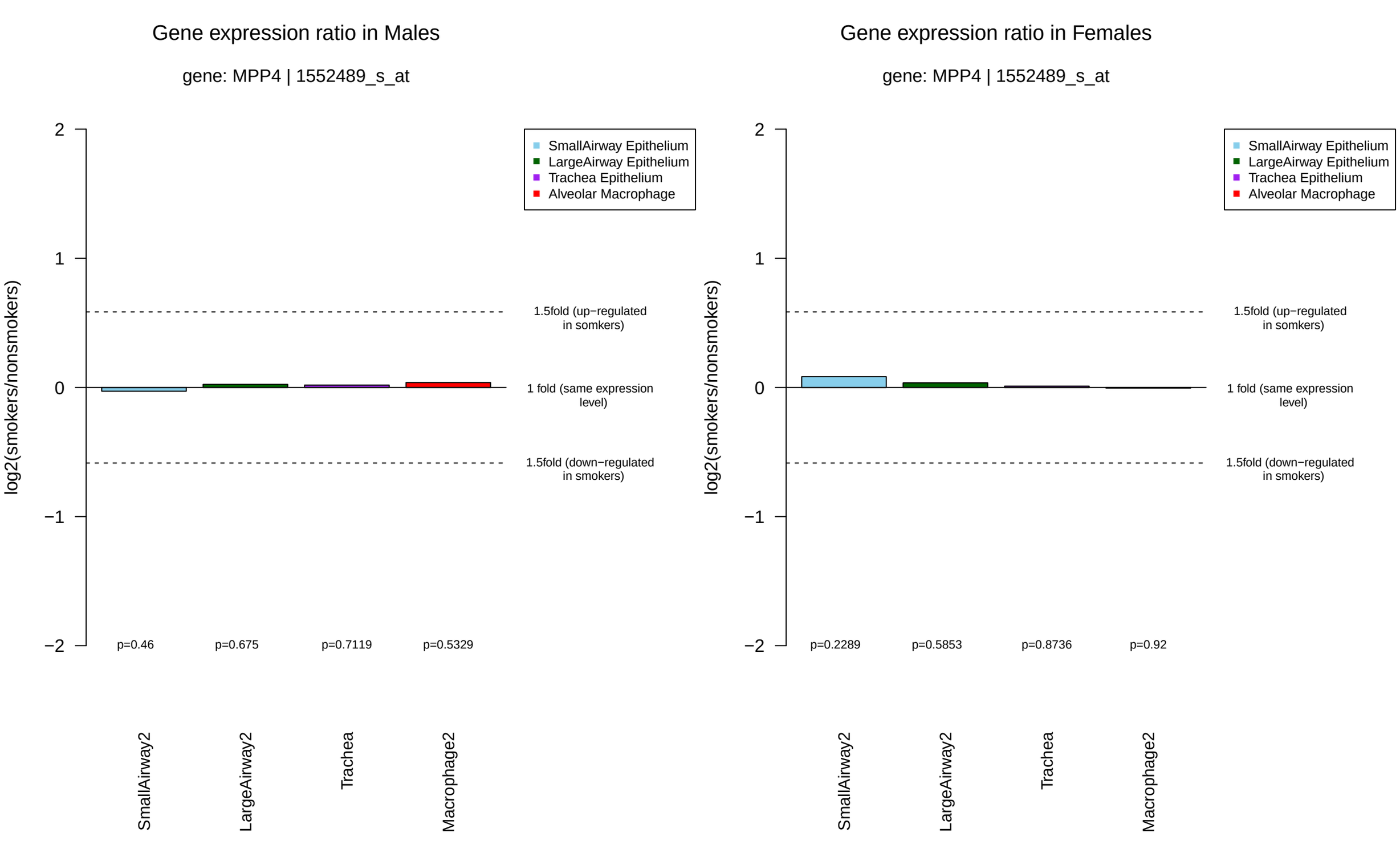
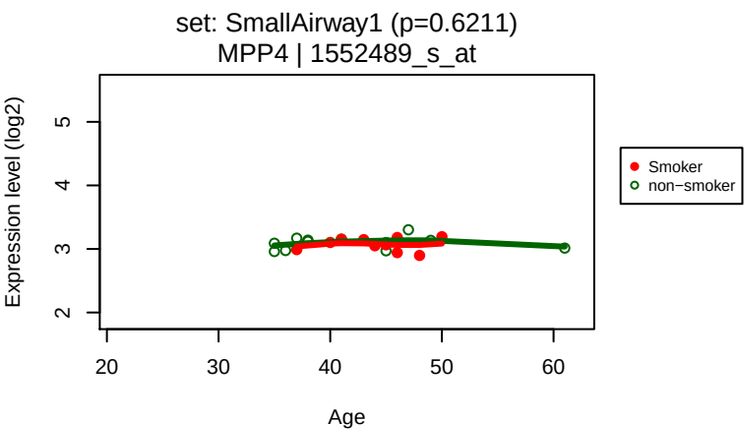
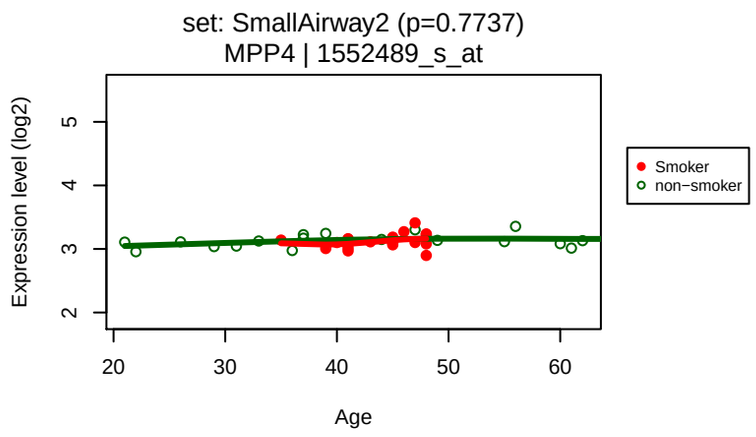
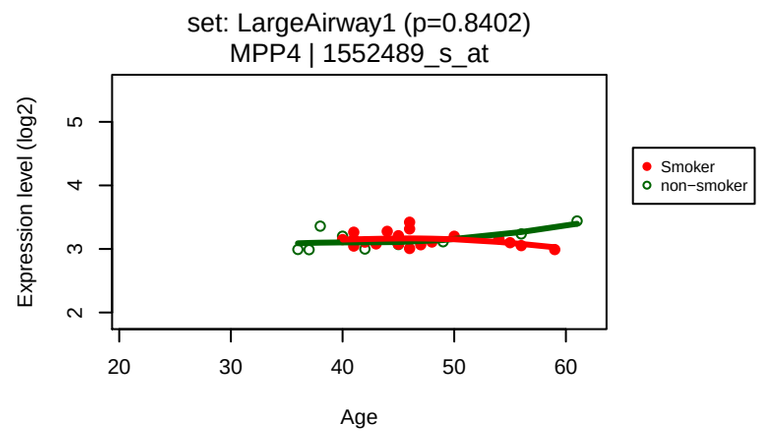

Gene expression of smokers and nonsmokers
gene: MPP4 | 1552489_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6211
p=0.7737
p=0.7406
p=0.8402
p=0.3975
p=0.5408
p=0.5996
p=0.6373
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

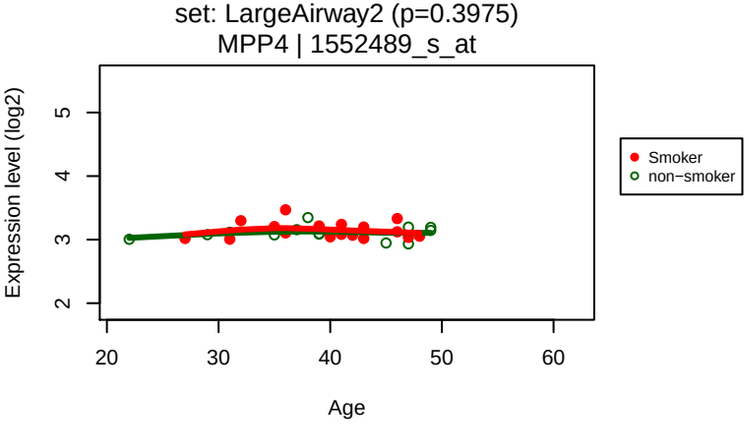

Gene expression ratio (smokers/nonsmokers)
gene: MPP4 | 1552489_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6211
p=0.7737
p=0.7406
p=0.8402
p=0.3975
p=0.5408
p=0.5996
p=0.6373
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

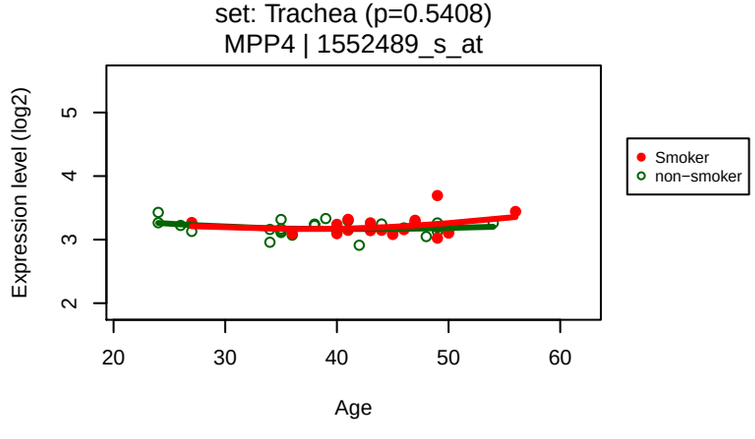

Gene expression ratio in Males
gene: MPP4 | 1552489_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.46
p=0.675
p=0.7119
p=0.5329
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPP4 | 1552489_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2289
p=0.5853
p=0.8736
p=0.92
SmallAirway2
LargeAirway2
Trachea
Macrophage2

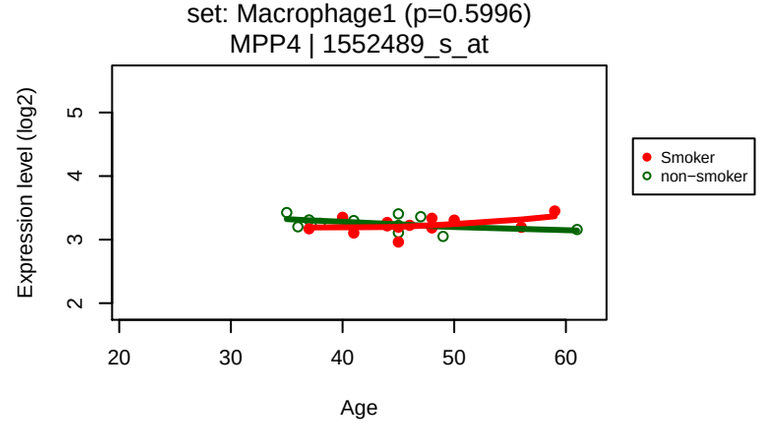

set: SmallAirway1 (p=0.6211)
MPP4 | 1552489_s_at
Expression level (log2)
Age
Smoker
non−smoker

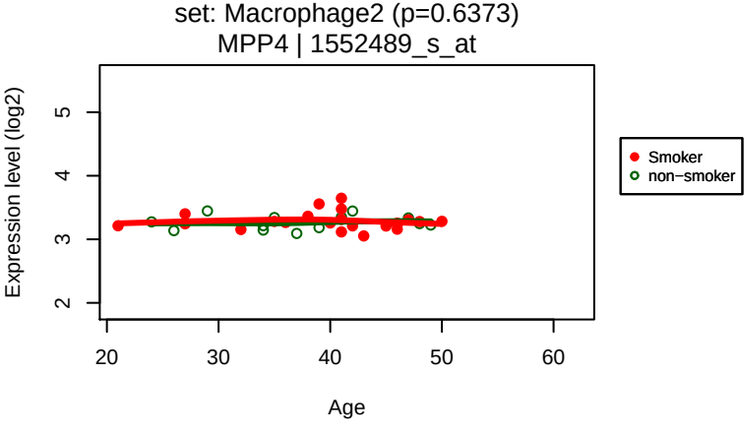

set: SmallAirway2 (p=0.7737)
MPP4 | 1552489_s_at
Expression level (log2)
Age
Smoker
non−smoker

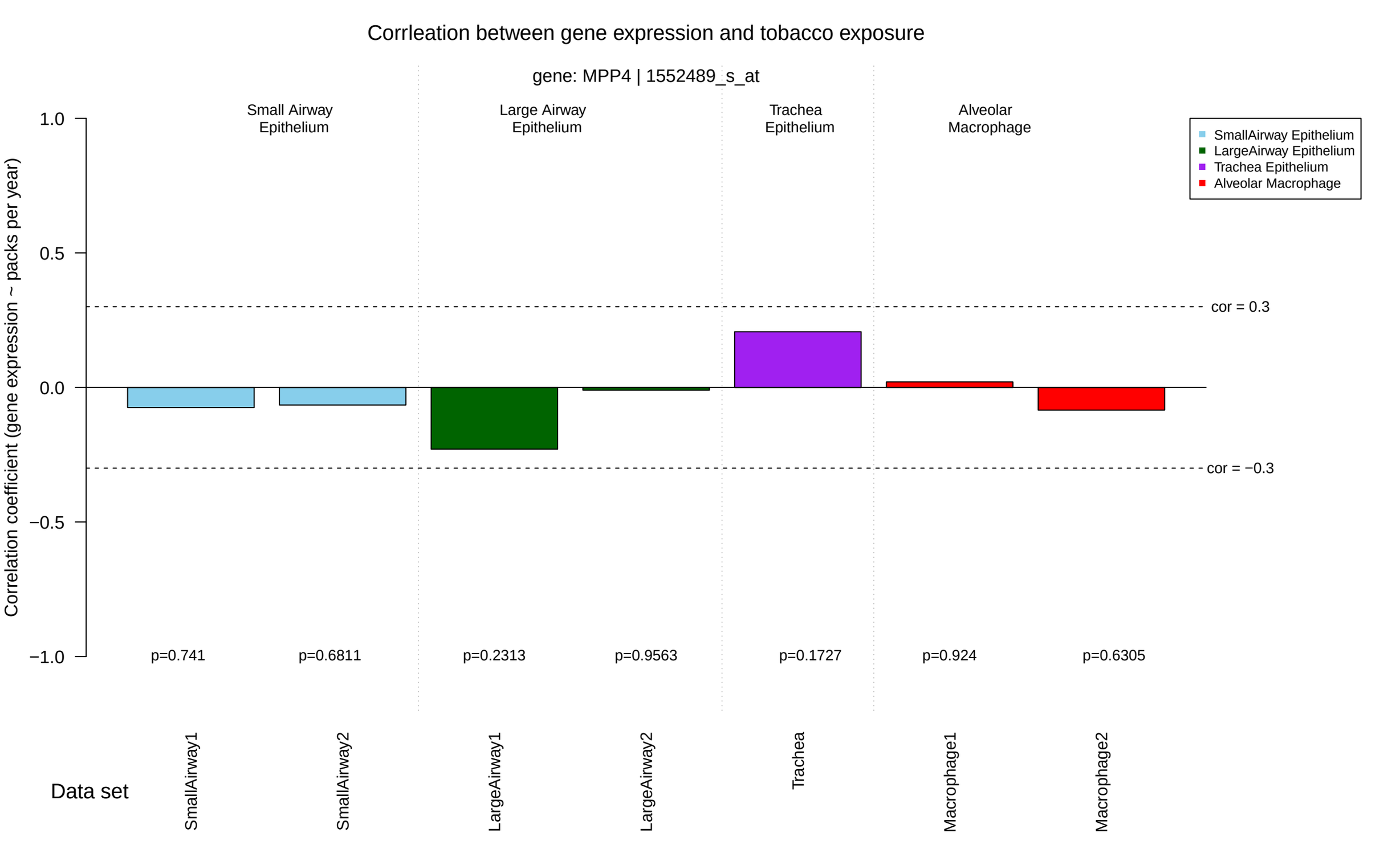

set: LargeAirway1 (p=0.8402)
MPP4 | 1552489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3975)
MPP4 | 1552489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5408)
MPP4 | 1552489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5996)
MPP4 | 1552489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6373)
MPP4 | 1552489_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPP4 | 1552489_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.741
p=0.6811
p=0.2313
p=0.9563
p=0.1727
p=0.924
p=0.6305
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2